

Solutions to MP363 Exam, Semester 1 '12-'13

(1)

P.1

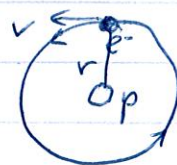
(a) Bohr's model is still based on the idea of an electron orbiting a proton, so it's based on a two-body problem of two particles bound together by the Coulomb force. Thus, if we assume that the electron orbits at velocity v in a circular orbit of radius r , we have

$$\frac{\mu v^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

where $\mu = \frac{m_e m_p}{m_e + m_p}$ is the reduced mass of the system. And since the electron is

assumed to be non relativistic, its energy is

$$E = \frac{1}{2} \mu v^2 - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r} = -\frac{1}{8\pi\epsilon_0} \frac{e^2}{r}.$$



Bohr's leap of logic was the assumption that the orbital angular momentum L of the electron was quantised, i.e. could only appear in discrete units. In particular, he said

$$L = \frac{h}{2\pi} n, \quad n=1,2,3,\dots$$

where h is Planck's constant. Since $L = \mu v r$, we can show

$$v = \frac{h}{2\pi \mu r} n$$

and thus

$$\frac{\mu v^2}{r} = \frac{h^2 n^2}{4\pi^2 \mu^2 r^3} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \Rightarrow r = \frac{\epsilon_0 h^2 n^2}{\pi \mu e^2}.$$

Putting this into the above expression for E therefore gives the answer:

$$E_n = -\frac{1}{8\pi\epsilon_0} \frac{e^2}{\left(\frac{\epsilon_0 h^2 n^2}{\pi \mu e^2}\right)} = -\frac{\mu e^4}{8\epsilon_0^2 h^2 n^2}.$$

We now know that this explanation isn't quite right, but it got the ball rolling in the right direction...

(b) The key to doing this part is to look at the reduced masses of the various two-particle systems and comparing them. The hydrogen atom has a reduced

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mass of

$$\mu^{(\text{hydrogen})} = \frac{m_e m_p}{m_e + m_p} = \frac{m_e \cdot 1836 m_e}{m_e + 1836 m_e} = \frac{1836}{1837} m_e.$$

Muonium, however, has

$$\mu^{(\text{muonium})} = \frac{m_e m_\mu}{m_e + m_\mu} = \frac{m_e \cdot 207 m_e}{m_e + 207 m_e} = \frac{207}{208} m_e.$$

Thus, the ratio of the energy levels are

$$\frac{E_n^{(\text{muonium})}}{E_n^{(\text{hydrogen})}} = \frac{\mu^{(\text{muonium})}}{\mu^{(\text{hydrogen})}} = 0.996.$$

Therefore, for a given n , the muonium energy is only a tiny bit less (about 0.4%) than the corresponding hydrogen level. In fact, to the accuracy that this formula for E_n holds, they're effectively the same.

The positronium reduced mass is

$$\mu^{(\text{positronium})} = \frac{m_e m_e}{m_e + m_e} = \frac{1}{2} m_e$$

so the energy ratio is

$$\frac{E_n^{(\text{positronium})}}{E_n^{(\text{hydrogen})}} = \frac{\mu^{(\text{positronium})}}{\mu^{(\text{hydrogen})}} = 0.5002,$$

so the positronium energy for a given n is a tiny bit more than half of the corresponding hydrogen energy.

P.2

(a) If $A: \mathcal{H} \rightarrow \mathcal{H}$ is an operator, then the associated adjoint operator $A^\dagger$ is that operator which satisfies the following identity for all vectors $\psi, \phi \in \mathcal{H}$:

$$\langle \psi | A^\dagger \phi \rangle = \langle A \psi | \phi \rangle.$$

(It's important to stress that this must hold for any choice of ψ and ϕ , not particular choices.)

Hermiticity of an operator - an important property that is required of observables - is simply defined as the property that an operator is equal to its own adjoint. Namely, A is hermitian if

$$A = A^\dagger \quad (\text{or, alternatively, } \langle \psi | A \phi \rangle = \langle A \psi | \phi \rangle \text{ for all } \psi, \phi).$$

③

(b) The key here is to show that $\langle \psi | H \phi \rangle = \langle H \psi | \phi \rangle$ for any two arbitrary states ψ and ϕ . In two cases,

we want

$$\int_{-\infty}^{\infty} \overline{\psi(x,t)} \left[-\frac{\hbar^2}{2m} \frac{\partial^2 \phi}{\partial x^2}(x,t) + V(x,t) \phi(x,t) \right] dx = \int_{-\infty}^{\infty} \overline{\left[-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}(x,t) + V(x,t) \psi(x,t) \right] \phi(x,t)} dx$$

First, the LHS:

$$\langle \psi | H \phi \rangle = \int_{-\infty}^{\infty} \bar{\psi} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \phi}{\partial x^2} + V \phi \right) dx$$

$$= -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \bar{\psi} \frac{\partial^2 \phi}{\partial x^2} dx + \int_{-\infty}^{\infty} \bar{\psi} V \phi dx$$

V is real, so $\bar{\psi} V \phi = (\bar{\psi} V) \phi = \overline{(V \psi)} \phi$. This is part of what we want, but we still need to look at the first term; note that

$$\bar{\psi} \frac{\partial^2 \phi}{\partial x^2} = \frac{\partial}{\partial x} \left(\bar{\psi} \frac{\partial \phi}{\partial x} \right) - \frac{\partial \bar{\psi}}{\partial x} \frac{\partial \phi}{\partial x}.$$

$$\int_{-\infty}^{\infty} \bar{\psi} \frac{\partial^2 \phi}{\partial x^2} dx = \int_{-\infty}^{\infty} \left[\frac{\partial}{\partial x} \left(\bar{\psi} \frac{\partial \phi}{\partial x} \right) - \frac{\partial \bar{\psi}}{\partial x} \frac{\partial \phi}{\partial x} \right] dx = \left[\bar{\psi} \frac{\partial \phi}{\partial x} \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{\partial \bar{\psi}}{\partial x} \frac{\partial \phi}{\partial x} dx.$$

Also,

$$\frac{\partial \bar{\psi}}{\partial x} \frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial \bar{\psi}}{\partial x} \phi \right) - \frac{\partial^2 \bar{\psi}}{\partial x^2} \phi, \text{ so}$$

$$\int_{-\infty}^{\infty} \bar{\psi} \frac{\partial^2 \phi}{\partial x^2} dx = \left[\bar{\psi} \frac{\partial \phi}{\partial x} \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \left[\frac{\partial}{\partial x} \left(\frac{\partial \bar{\psi}}{\partial x} \phi \right) - \frac{\partial^2 \bar{\psi}}{\partial x^2} \phi \right] dx$$

$$= \left[\bar{\psi} \frac{\partial \phi}{\partial x} - \frac{\partial \bar{\psi}}{\partial x} \phi \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{\partial^2 \bar{\psi}}{\partial x^2} \phi dx.$$

Now, any physical states must be normalisable, i.e. $\|\psi\| = 1$.

This means that $\|\psi\|^2 = \langle \psi | \psi \rangle = \int_{-\infty}^{\infty} |\psi(x,t)|^2 dx$ must exist.

This can only happen if $|\psi(x,t)|^2 \rightarrow 0$ as $x \rightarrow \pm\infty$, and

this, in turn, can only happen if $\boxed{|\psi(x,t)| \text{ itself vanishes as } x \rightarrow \pm\infty}$

Thus, provided $\frac{\partial \psi}{\partial x}$ and $\frac{\partial \phi}{\partial x}$ are well-behaved as $x \rightarrow \pm\infty$, then

$$\left[\bar{\psi} \frac{\partial \phi}{\partial x} - \frac{\partial \bar{\psi}}{\partial x} \phi \right]_{-\infty}^{\infty} = 0. \text{ Therefore, we conclude that}$$

$$\langle \psi | H \phi \rangle = -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \frac{\partial^2 \bar{\psi}}{\partial x^2} \phi dx + \int_{-\infty}^{\infty} (V \bar{\psi}) \phi dx.$$

x is a real variable, and $\hbar$ and m are real constants, so

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \bar{\psi}}{\partial x^2} = \overline{\left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} \right)}, \text{ giving}$$

$$\langle \psi | H \phi \rangle = \int_{-\infty}^{\infty} \overline{\left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} \right)} \phi dx + \int_{-\infty}^{\infty} (V \bar{\psi}) \phi dx$$

$$= \int_{-\infty}^{\infty} \overline{\left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V \psi \right)} \phi dx$$

(d)

$= \langle H\psi | \psi \rangle$
 proving $\boxed{H \text{ is hermitian.}}$

P.3

(a) This can be done the really easy way or the kind of easy way: the really easy way is to recall that if $|\psi_n\rangle$ are energy eigenstates, and a state has the form $|\psi\rangle = \sum \alpha_n |\psi_n\rangle$, then the expectation value of the Hamiltonian is

$$\langle H \rangle_\psi = \sum_n |\alpha_n|^2 E_n.$$

In the case we have here, $|\psi_n\rangle = |e_n\rangle$ and the only three nonzero coefficients are $\alpha_0 = \frac{1}{\sqrt{6}}$, $\alpha_1 = -\frac{1}{\sqrt{2}}$ and $\alpha_2 = \frac{1}{\sqrt{3}}$. Therefore,

$$\begin{aligned} \langle H \rangle_\psi &= \left| \frac{1}{\sqrt{6}} \right|^2 E_0 + \left| -\frac{1}{\sqrt{2}} \right|^2 E_1 + \left| \frac{1}{\sqrt{3}} \right|^2 E_2 \\ &= \frac{1}{6} \left(\frac{\hbar\omega}{2} \right) + \frac{1}{2} \left(\frac{3\hbar\omega}{2} \right) + \frac{1}{3} \left(\frac{5\hbar\omega}{2} \right) = \boxed{\frac{5}{3} \hbar\omega}. \end{aligned}$$

The kind of easy way is using $\langle H \rangle_\psi = \langle \psi | H | \psi \rangle$.

$$\begin{aligned} H|\psi\rangle &= \frac{1}{\sqrt{6}} H|e_0\rangle + \frac{1}{\sqrt{2}} H|e_1\rangle + \frac{1}{\sqrt{3}} H|e_2\rangle \\ &= \frac{1}{\sqrt{6}} E_0 |e_0\rangle - \frac{1}{\sqrt{2}} E_1 |e_1\rangle + \frac{1}{\sqrt{3}} E_2 |e_2\rangle \end{aligned}$$

so

$$\begin{aligned} \langle H \rangle_\psi &= \left(\frac{1}{\sqrt{6}} \langle e_0| - \frac{1}{\sqrt{2}} \langle e_1| + \frac{1}{\sqrt{3}} \langle e_2| \right) \left(\frac{E_0}{\sqrt{6}} |e_0\rangle - \frac{1}{\sqrt{2}} E_1 |e_1\rangle + \frac{1}{\sqrt{3}} E_2 |e_2\rangle \right) \\ &= \frac{E_0}{6} \langle e_0|e_0\rangle - \frac{E_0}{\sqrt{12}} \langle e_1|e_0\rangle + \frac{E_0}{\sqrt{18}} \langle e_2|e_0\rangle - \frac{E_1}{\sqrt{12}} \langle e_0|e_1\rangle \\ &\quad + \frac{E_1}{2} \langle e_1|e_1\rangle - \frac{E_1}{\sqrt{6}} \langle e_2|e_1\rangle + \frac{E_2}{\sqrt{18}} \langle e_0|e_2\rangle - \frac{E_2}{\sqrt{6}} \langle e_1|e_2\rangle \\ &\quad + \frac{E_2}{3} \langle e_2|e_2\rangle \\ &= \frac{E_0}{6} + \frac{E_1}{2} + \frac{E_2}{3} \end{aligned}$$

because $\langle e_i | e_j \rangle = \delta_{ij}$. This also gives $\boxed{\frac{5}{3} \hbar\omega}$.

(b) In terms of the raising and lowering operators,

$$X = \frac{i}{2m\omega} (a_- - a_+), \quad P = \frac{1}{2} (a_- + a_+).$$

Since both $\langle X \rangle_\psi$ and $\langle P \rangle_\psi$ will be combinations of $\langle a_\pm \rangle_\psi$, let's compute those first.

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$$\begin{aligned}
 a_+ |\psi\rangle &= \frac{1}{\sqrt{6}} a_+ |e_0\rangle + \frac{1}{\sqrt{2}} a_+ |e_1\rangle + \frac{1}{\sqrt{3}} a_+ |e_2\rangle \\
 &= \frac{1}{\sqrt{6}} (\sqrt{2m\hbar\omega} |e_1\rangle) - \frac{1}{\sqrt{2}} (\sqrt{4m\hbar\omega} |e_2\rangle) + \frac{1}{\sqrt{3}} (\sqrt{6m\hbar\omega} |e_3\rangle) \\
 &= \sqrt{\frac{m\hbar\omega}{3}} |e_1\rangle - \sqrt{2m\hbar\omega} |e_2\rangle + \sqrt{2m\hbar\omega} |e_3\rangle
 \end{aligned}$$

so

$$\begin{aligned}
 \langle a_+ \rangle_\psi &= \left(\frac{1}{\sqrt{6}} \langle e_0| - \frac{1}{\sqrt{2}} \langle e_1| + \frac{1}{\sqrt{3}} \langle e_2| \right) \left(\sqrt{\frac{m\hbar\omega}{3}} |e_1\rangle - \sqrt{2m\hbar\omega} |e_2\rangle + \sqrt{2m\hbar\omega} |e_3\rangle \right) \\
 &= -\sqrt{\frac{m\hbar\omega}{6}} + \sqrt{\frac{2m\hbar\omega}{3}} = -\sqrt{\frac{3m\hbar\omega}{2}}.
 \end{aligned}$$

$$\begin{aligned}
 a_- |\psi\rangle &= \frac{1}{\sqrt{6}} a_- |e_0\rangle - \frac{1}{\sqrt{2}} a_- |e_1\rangle + \frac{1}{\sqrt{3}} a_- |e_2\rangle \\
 &= \frac{1}{\sqrt{6}} (0) - \frac{1}{\sqrt{2}} (\sqrt{2m\hbar\omega} |e_0\rangle) + \frac{1}{\sqrt{3}} (\sqrt{4m\hbar\omega} |e_1\rangle) \\
 &= -\sqrt{m\hbar\omega} |e_0\rangle + \sqrt{\frac{4m\hbar\omega}{3}} |e_1\rangle
 \end{aligned}$$

so

$$\begin{aligned}
 \langle a_- \rangle_\psi &= \left(\frac{1}{\sqrt{6}} \langle e_0| - \frac{1}{\sqrt{2}} \langle e_1| + \frac{1}{\sqrt{3}} \langle e_2| \right) \left(-\sqrt{m\hbar\omega} |e_0\rangle + \sqrt{\frac{4m\hbar\omega}{3}} |e_1\rangle \right) \\
 &= -\sqrt{\frac{m\hbar\omega}{6}} - \sqrt{\frac{2m\hbar\omega}{3}} = -\sqrt{\frac{3m\hbar\omega}{2}}.
 \end{aligned}$$

Therefore,

$$\langle X \rangle_\psi = \frac{i}{2m\omega} \langle a_- \rangle_\psi - \langle a_+ \rangle_\psi = \boxed{0},$$

$$\langle P \rangle_\psi = \frac{1}{2} (\langle a_- \rangle_\psi + \langle a_+ \rangle_\psi) = \boxed{-\sqrt{\frac{3m\hbar\omega}{2}}}.$$

A measurement of the particle's position will therefore average out to it being right in the middle, whereas its momentum, on the average, is to the left.